



# **National Textile University**

*Department of Computer Science*

**Subject:**

Mobile application and development

**Submitted To:**

Sir Rehan Sarwar

**Submitted By:**

Noor ul Ain

**Registration No:**

23-NTU-CS-1221

**Lab no:**

7

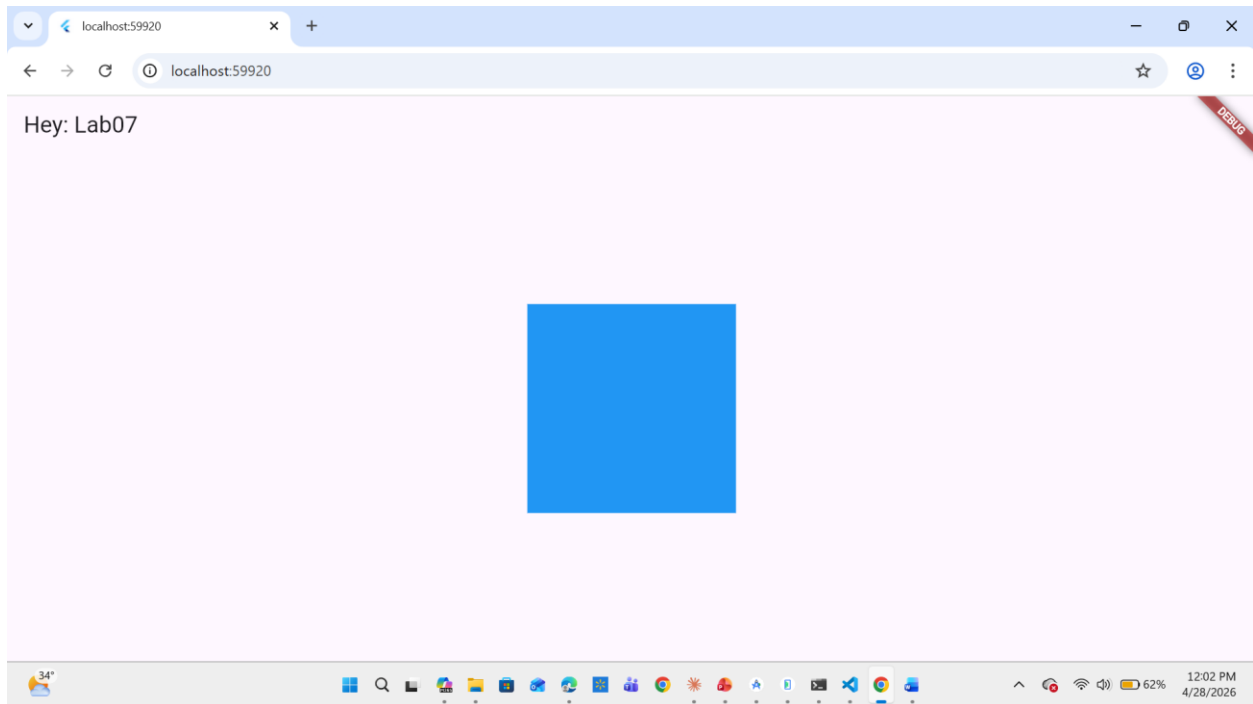
**Semester:**

6<sup>th</sup>

## PHASE 1: GESTUREDETECTOR

```
1 import 'dart:math';
2 import 'package:flutter/material.dart';
3 void main() {
4   runApp(MaterialApp(
5     home: Lab07(),
6   ));
7 }
8 class Lab07 extends StatefulWidget {
9   const Lab07({super.key});
10
11   @override
12   State<Lab07> createState() => _Lab07State();
13 }
14 class _Lab07State extends State<Lab07> {
15
16   Color boxColor = Colors.blue;
17   double borderRadius = 0;
18
19   final Color originalColor = Colors.blue;
20
21   @override
22   Widget build(BuildContext context) {
23     return Scaffold(
24       appBar: AppBar(
25         title: Text("Hey: Lab07"),
26       ),
27       body: Center(
28         child: GestureDetector(
29           onTap: () {
30             setState(() {
31               final random = Random();
32               boxColor = Color.fromARGB(
33                 255,
34                 random.nextInt(256),
35                 random.nextInt(256),
36                 random.nextInt(256),
37               );
38             });
39           },
40           onDoubleTap: () {
41             setState(() {
42               borderRadius = borderRadius == 0 ? 100 : 0;
43             });
44           },
45           onLongPress: () {
46             setState(() {
47               boxColor = originalColor;
48               borderRadius = 0;
49             });
50           },
51           ScaffoldMessenger.of(context).showSnackBar(
52             SnackBar(content: Text("Resetting...")),
53           ),
54         ),
55         child: AnimatedContainer(
56           duration: Duration(milliseconds: 300),
57           height: 200,
58           width: 200,
59           decoration: BoxDecoration(
60             color: boxColor,
61             borderRadius: BorderRadius.circular(borderRadius),
62           ),
63         ),
64       ),
65     );
66   }
67 }
68 }
```

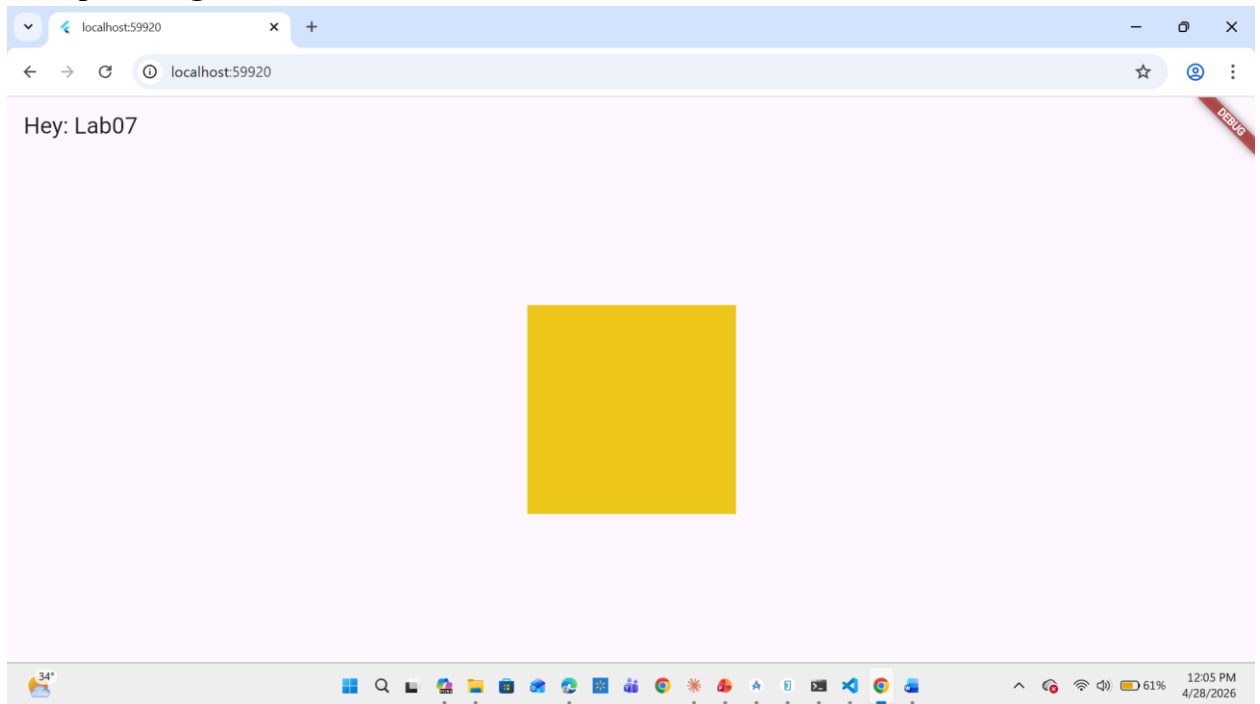
### Task 1.1: Create a Container (200x200) with a unique background color.



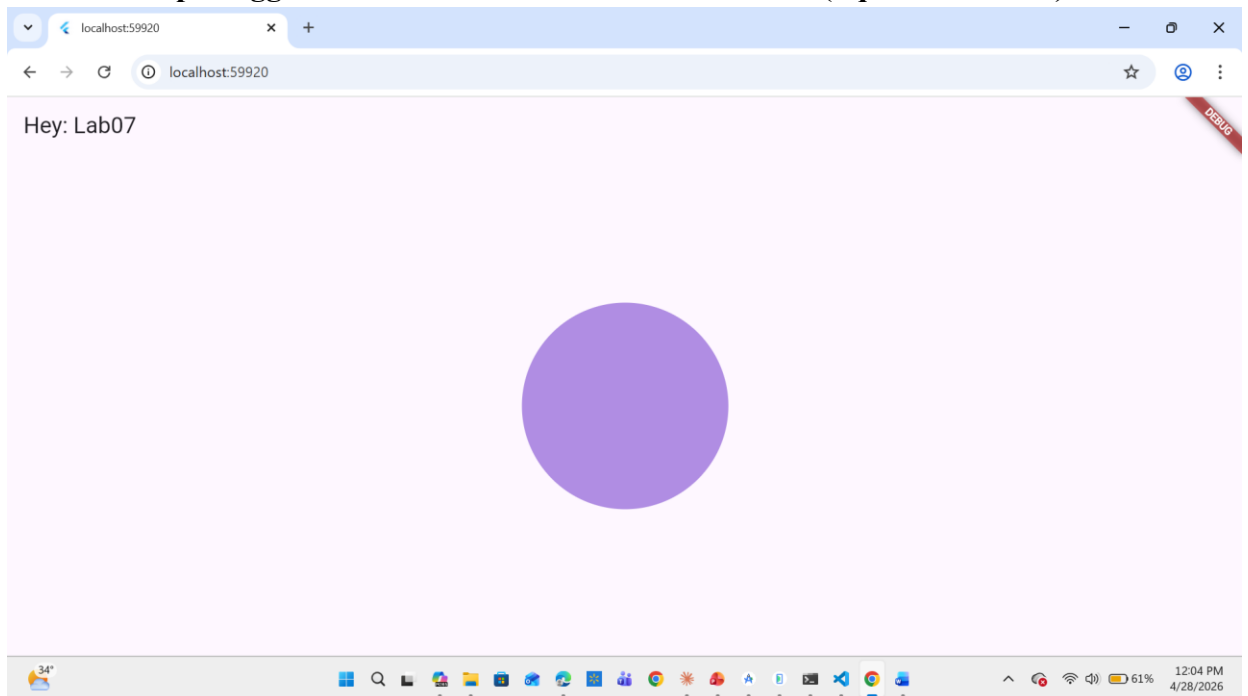
### Task 1.2: Wrap the container in a GestureDetector.

### Task 1.3: Logic Implementation:

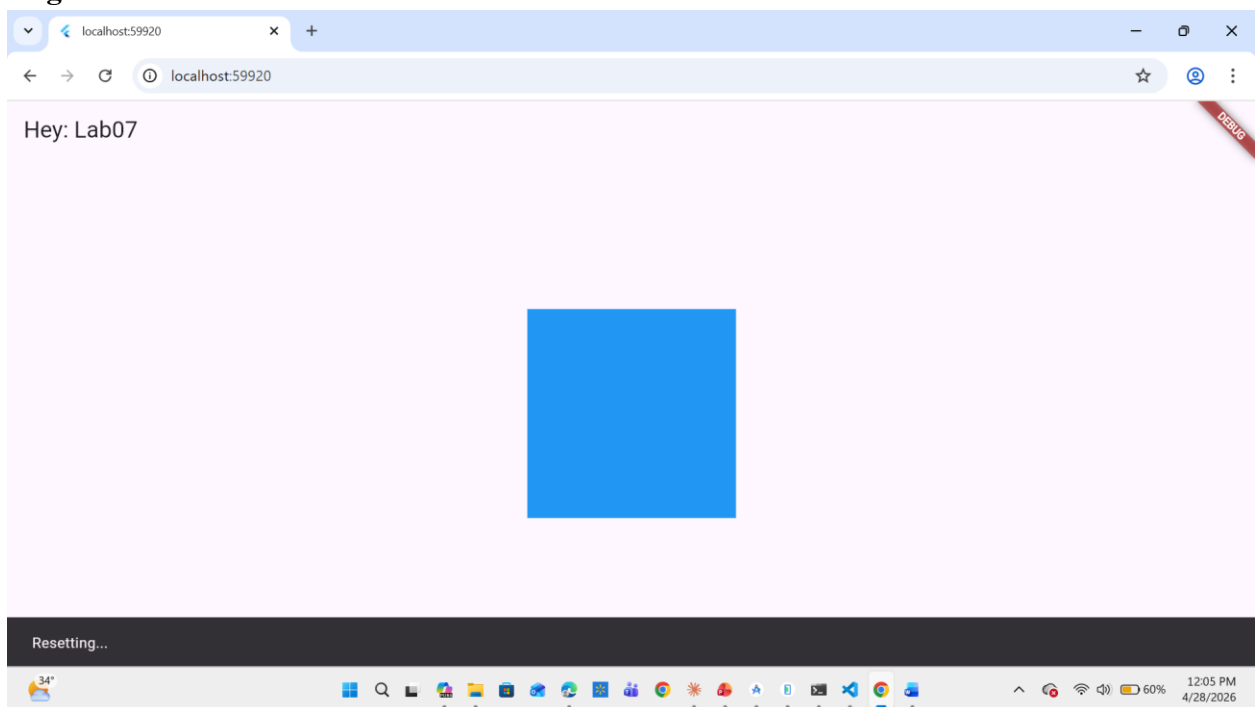
- **onTap: Change the container's color to a random Colors value.**



- **onDoubleTap: Toggle the borderRadius between 0 and 100 (Square to Circle).**



- **onLongPress: Show a SnackBar saying "Resetting..." and return the box to its original state.**



## PHASE 2: SLIDERS

Task 2.1: Create a vertical Column containing a Text widget displaying a "Value" variable.

```
1 //Task 2.1
2 Text("Value: ${_currentSliderValue.round()}"),
3
```

Task 2.2: Add a Slider widget:

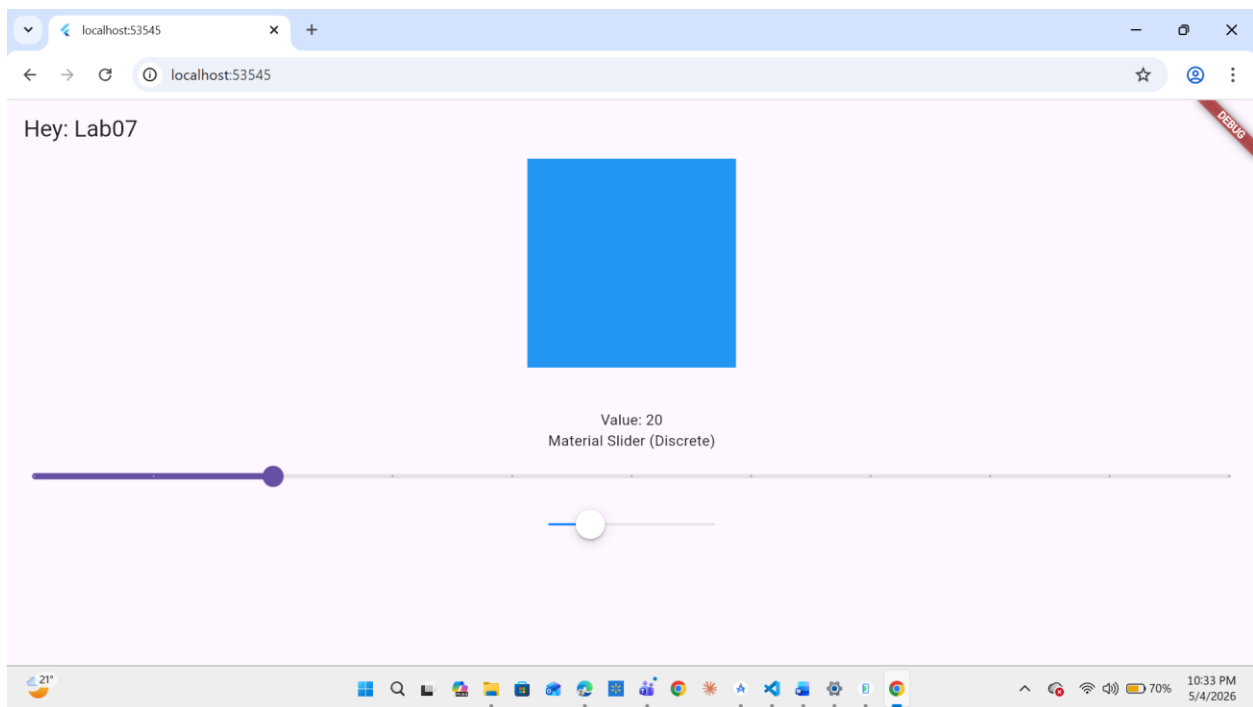
- Set min to 0.0 and max to 100.
- Set divisions to 10 to create discrete steps.
- Update the "Value" text in real-time within onChanged.

```
1 const Text("Material Slider (Discrete)"),
2     Slider(
3       value: _currentSliderValue,
4       min: 0,
5       max: 100,
6       divisions: 10,
7       label: _currentSliderValue.round().toString(),
8       onChanged: (double value) {
9         setState(() {
10           _currentSliderValue = value;
11         });
12       },
13     ),
```

Task 2.3: Implement a CupertinoSlider below the Material one to observe the difference in native "look and feel".

```
1 //Task 2.3
2 CupertinoSlider(
3   key:const Key("cupertinp_slider"),
4   value:_currentSliderValue,
5   min:0,
6   max:100,
7   activeColor: CupertinoColors.activeBlue,
8   onChanged: (double value) {
9     setState(() {
10      _currentSliderValue = value; // Update the shared state
11    });
12  },)
```

## PHASE 2 OUTPUT:



## PHASE 3: THE "MOOD & COLOR" MIXER

The Challenge: Build a mini-app on a single screen where:



```
1  class _Lab07State extends State<Lab07> {  
2    // Phase 3: RGB State Variables  
3    double _red = 33;  
4    double _green = 150;  
5    double _blue = 243;  
6  
7  
8    double _scaleFactor = 1.0;
```

1. The Challenge: Build a mini-app on a single screen where: 1. Three Sliders control the Red, Green, and Blue channels of a preview box.



```
1  // Task 3.1: RGB Sliders  
2    _buildColorSlider("Red", _red, Colors.red, (val) {  
3      setState(() => _red = val);  
4    }),  
5    _buildColorSlider("Green", _green, Colors.green, (val) {  
6      setState(() => _green = val);  
7    }),  
8    _buildColorSlider("Blue", _blue, Colors.blue, (val) {  
9      setState(() => _blue = val);  
10   }));
```

2. A GestureDetector on the preview box allows the user to Long Press to "Copy" the Hex code to the console

```
1 // Phase 3.2: Long Press to "Copy" Hex
2
3   onLongPress: () {
4     String hexCode = '#${_boxColor.value.toRadixString(16).padLeft(8, '0').toUpperCase().substring(2)}';
5     print("Hex Code: $hexCode"); // Prints to console[cite: 1]
6     ScaffoldMessenger.of(context).showSnackBar(
7       SnackBar(content: Text("Copied Hex: $hexCode to console")),
8     );
9   },
```

3. Horizontal Drag (onHorizontalDragUpdate): Should scale the size of the preview box up or down.

```
1 // Phase 3.3: Horizontal Drag to Scale
2
3   onHorizontalDragUpdate: (details) {
4     setState(() {
5       // Update scale based on drag movement
6       _scaleFactor += details.delta.dx * 0.01;
7       // Clamp to keep the box within visible limits
8       _scaleFactor = _scaleFactor.clamp(0.5, 2.5);
9     });
10  },
```



## PHASE 3 OUTPUT:

